PROGRESS REPORT 1 · GAME-BASED LEARNING

XP Arena

A live, Kahoot-style quiz on medical prefixes, suffixes and roots. Every correct answer pays out twice: XP that counts toward the class grade, and coins that buy avatar cosmetics.

RECALL · DECODE · APPLY · HIGH-FIDELITY PROTOTYPE

1 · PROJECT TITLE / GBL

XP Arena: A Live Medical-Terminology Quiz Game

Game-based learning idea for mastering medical word parts. Students join a teacher-hosted room with a 6-digit PIN, answer the same timed question on their own phones, see each term broken into its parts after every question, and spend earned coins on their avatar.

2 · GROUP MEMBERS PRESENT TODAY & ROLES

Lasala	• **Task 1** · Reviewed the pitch deck and chose the tech stack • **Task 2** · Built the live game server and the scoring rules
Ausan	• **Task 3** · Wrote the question bank • **Task 4** · Built the teacher's projector screens
David	• **Task 7** · Overhauled the UI/UX (redesign) • **Task 8** · Tested a full round and fixed what we found
Mayam	• **Task 5** · Built the student's phone screens • **Task 6** · Built the Locker & Coin Shop and the class record
Jomoc	*Absent today*

3 · DATE OF PROGRESS

Thursday, October 22, 2026

Status: working prototype, tested end to end

4 Tasks Completed Today

We turned the *XP Arena – Design a GBL Idea* pitch deck into a working, high-fidelity prototype. One computer hosts the game on the projector, and students play from their phones over the classroom Wi-Fi. Each task below is followed by screenshots taken from a full test round with one student and five simulated classmates.

Task 1 Reviewed the pitch deck and chose the tech stack

L Lasala DONE

We went through all 9 slides (learning objectives, target learners, game concept, session flow, mechanics, rewards, expected outcomes) and turned each one into a feature list. Stack: **Node.js + TypeScript** throughout, **React + Vite** for the screens, **Express** for profiles, shop and records, and **Socket.IO** for live rooms. Shared rule files keep the server and the screens in agreement on scoring and shop items.

Task 2 Built the live game server and the scoring rules

L Lasala DONE

- **Rooms:** a 6-digit PIN and QR code per room; players reconnect automatically if their phone drops; "demo players" let one person test a full round.
- **Fair timing:** the server owns the 20-second clock and computes every score, so a phone cannot fake a fast answer.
- **Mechanics from slide 05:** up to 1,000 XP per correct answer (faster = more), a +200 streak bonus from 3 in a row that keeps stacking, +1 grade bonus point per 1,000 XP (capped at +5 per term), 10 coins per correct answer and +25 for the top 3, and a top-5 leaderboard after every question.
- **Checked by 6 automated tests** on the scoring rules; all pass.

Task 3 Wrote the question bank

A Ausan DONE

38 questions on prefixes, suffixes and roots: 16 **recall** (what does *nephr-* mean?), 12 **decode** (break down *gastritis*) and 10 **apply** (pick the right term for a patient scenario). Each round has 15 questions that ramp from recall to apply. The last question is an untimed review with no speed bonus, following the trade-off noted on slide 07.

Task 4 Built the teacher's projector screens

A Ausan DONE

Lobby with the PIN, join address and QR code; the live question with a countdown ring and answer count; the answer reveal with how the class voted, the term broken into its parts, and the top 5; and a final podium.

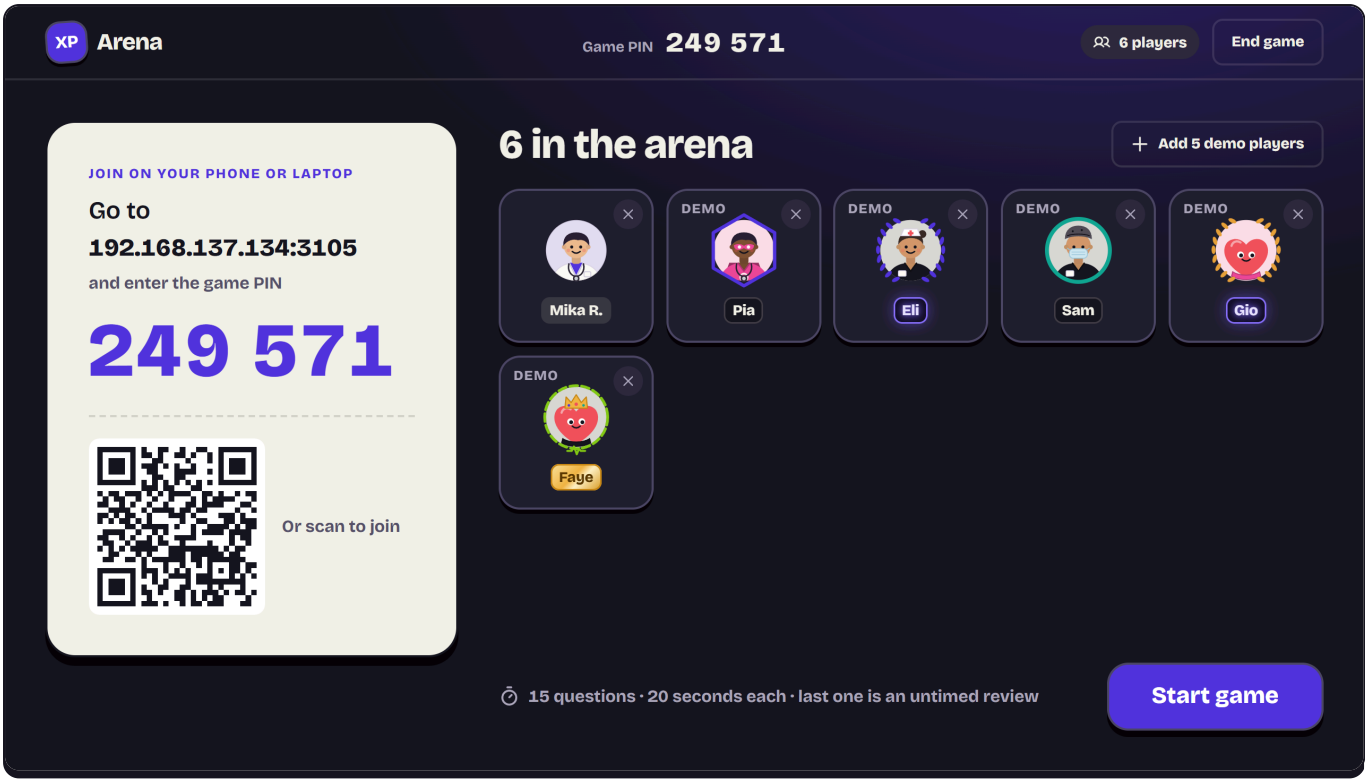


Figure 1. Projector lobby: game PIN, join address and QR code on the left; joined players (one student, five demo players) on the right.

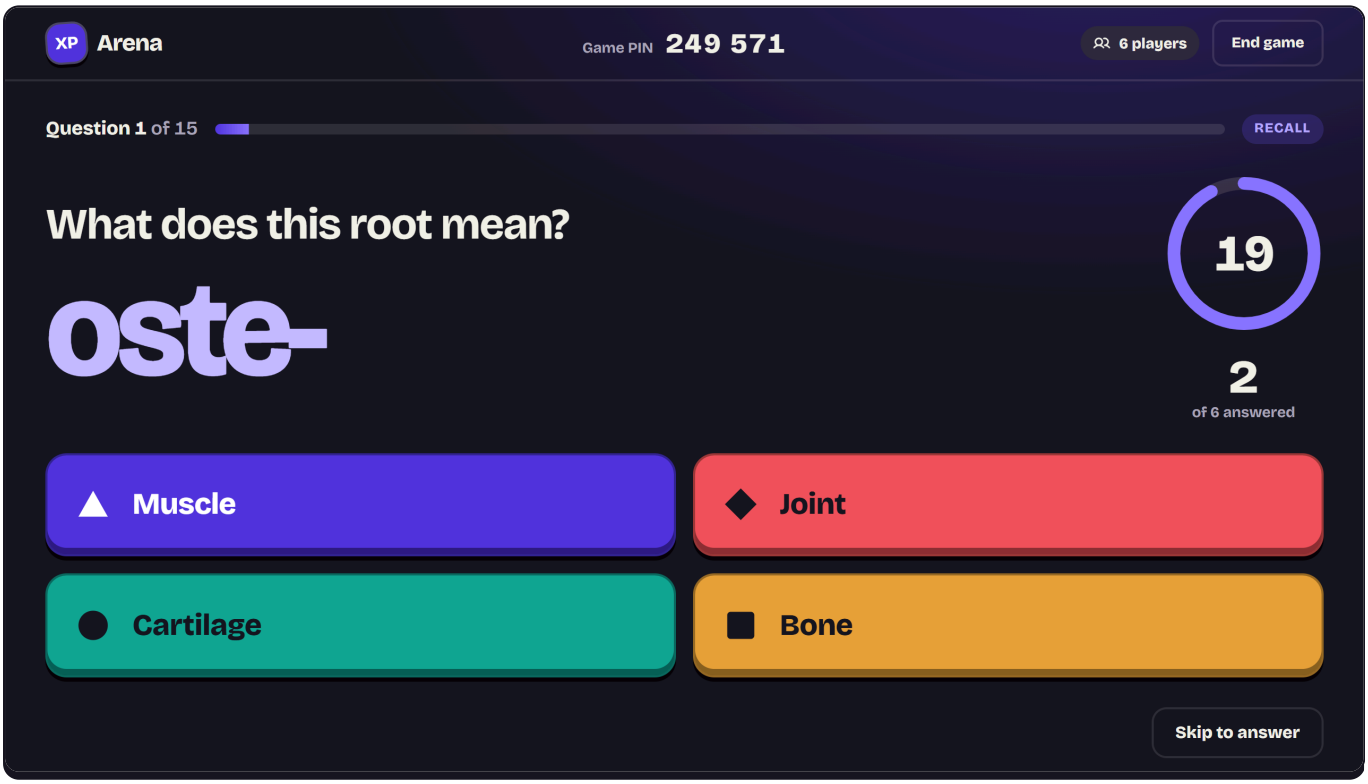


Figure 2. A live question on the projector: 20-second countdown ring, number of players who have answered, and four color-and-shape answer blocks.

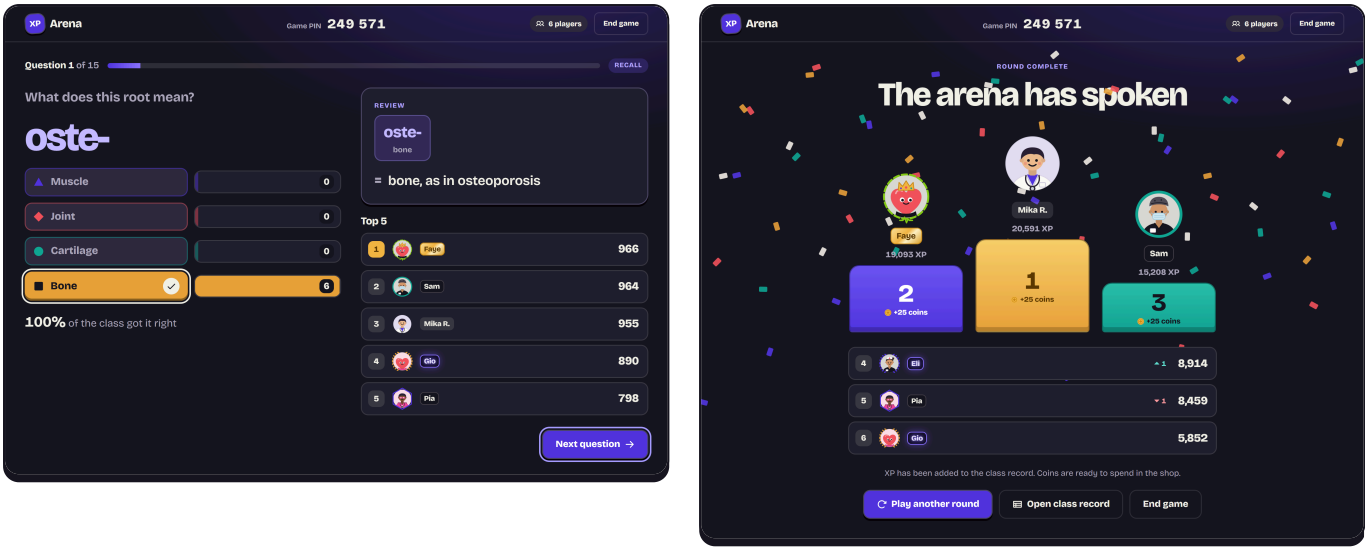


Figure 3. Left: answer reveal (the Review step). The correct answer gets a ring and check, the bars show how the class voted, the term is broken into its parts (*oste-* = bone), and the top 5 updates. Right: end-of-round podium for the top 3 (+25 coins each) with the rest of the ranking.

Task 5 Built the student's phone screens

M Mayam DONE

Create a player, join by QR or PIN, choose a loadout in the lobby, answer on big color-and-shape buttons, then see the result: speed points, streak bonus, coins earned, current rank and the term breakdown. The round ends with a personal summary and progress toward the +5 grade bonus.

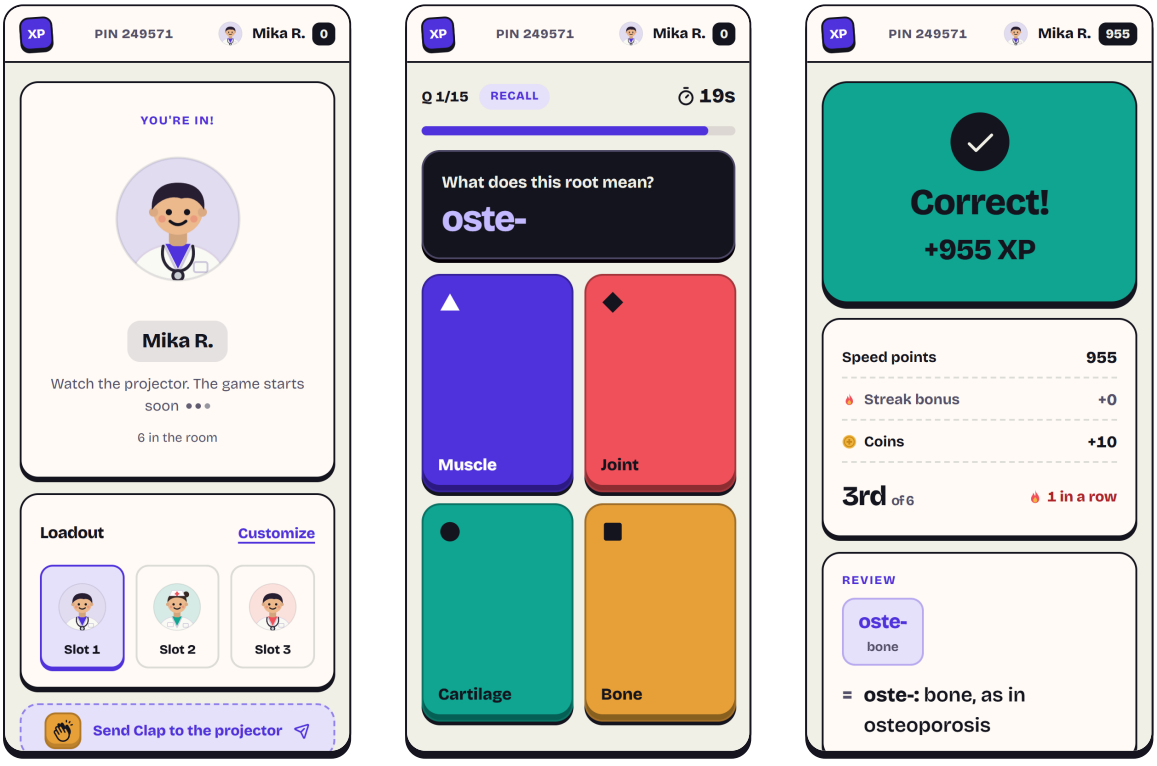


Figure 4. Phone screens, left to right: lobby with loadout slots, answer pad during a question, and the result after a correct answer (+955 XP).

Task 6 Built the Locker & Coin Shop and the class record

M Mayam DONE

- **Locker & Coin Shop (slide 06):** 6 avatars (doctor, nurse, lab tech and 3 mascots: heart, brain and bone), skins, profile frames, nameplates, answer-card themes and victory emotes that float up on the projector. Players pick outfit, frame and skin-tone colors and save 3 loadouts. Seasonal items unlock with answer streaks instead of coins. Coins only buy looks, never grades.

- **Class record (teacher view):** each student's XP for the term, grade-bonus progress, accuracy against the 80% target from slide 07, and a "start a new term" reset.

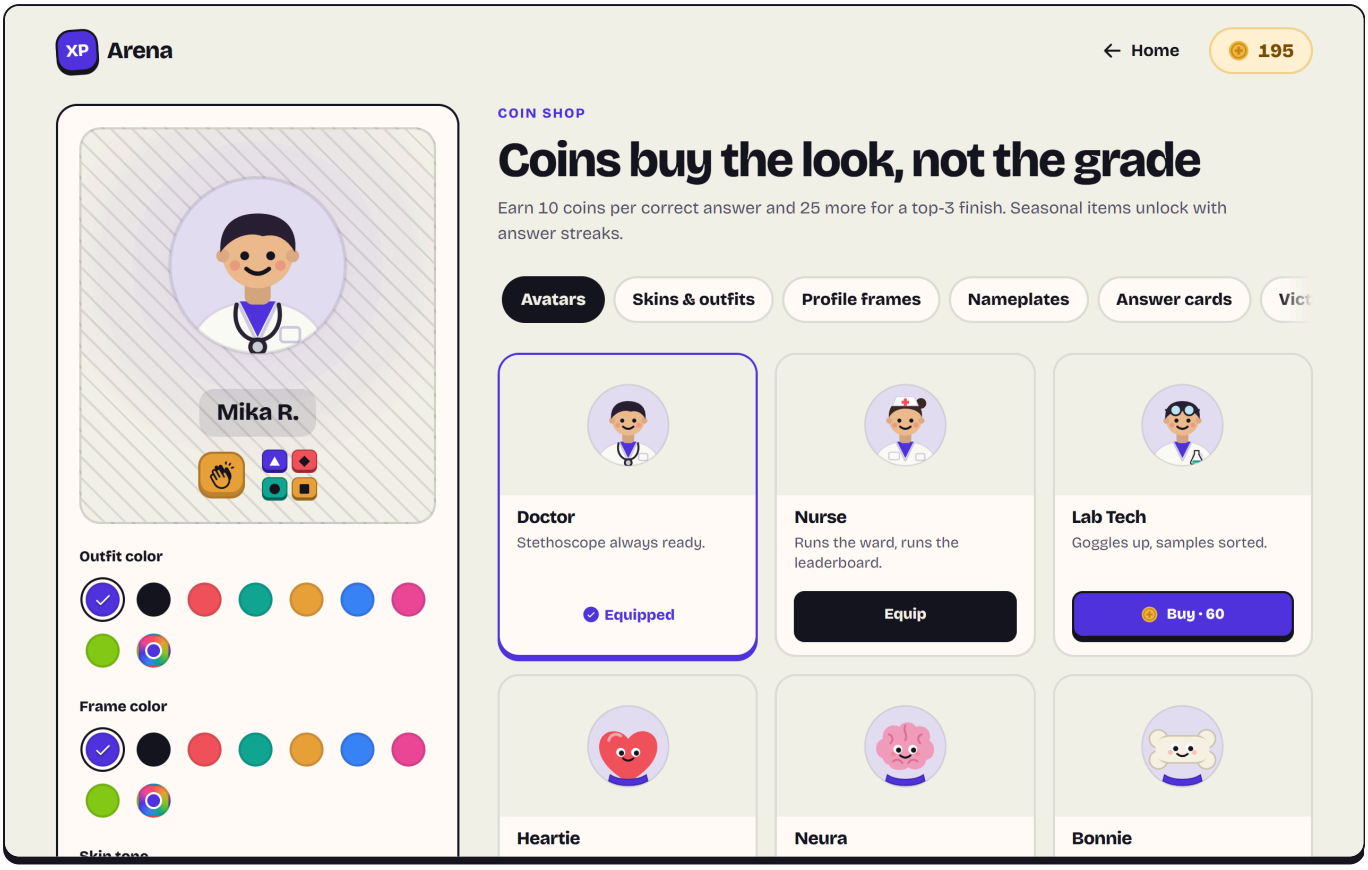


Figure 5. Locker & Coin Shop on a laptop: live avatar preview with color pickers and loadouts on the left, shop items by category on the right.

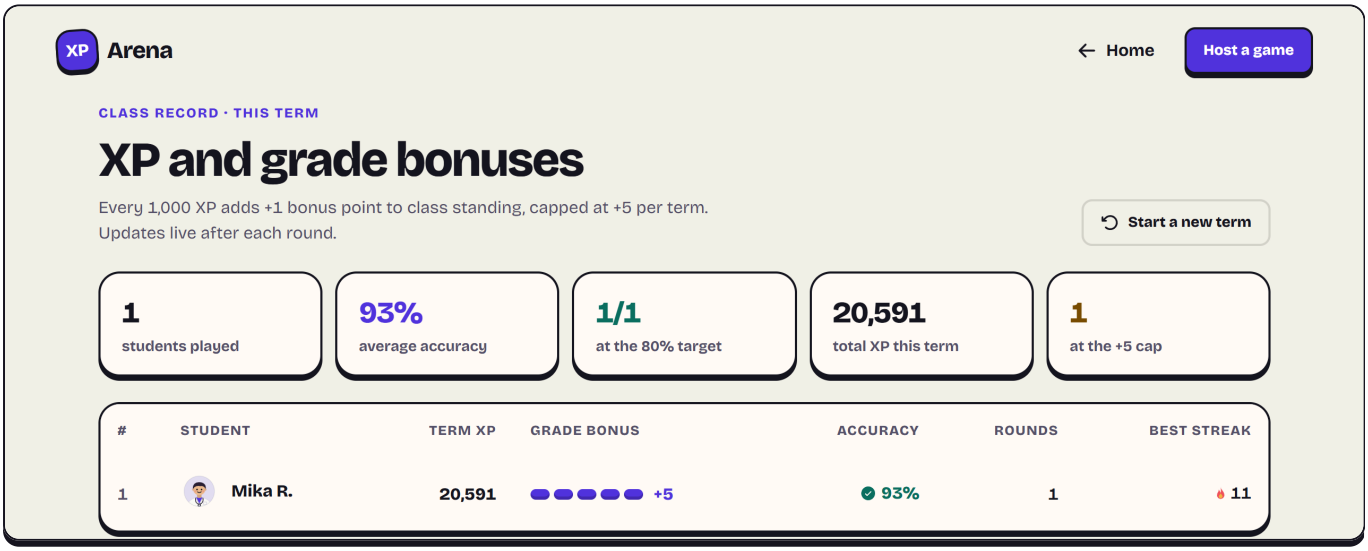


Figure 6. Class record after the test round: term XP, grade-bonus pips (+5 reached), accuracy (93%, above the 80% target), rounds played and best streak.

Task 7 Overhauled the UI/UX with the ui-ux-pro-max design skill

David DONE

- **One design system:** kept the deck's ink, cream and violet colors and Bricolage Grotesque font, restyled as the skill's recommended "vibrant, block-based" quiz-game look with chunky buttons and color blocks.
- **Dark mode** on phone screens, following the device setting. The projector is always dark.

- **Readable colors:** every text and background pair now meets the 4.5:1 contrast standard (checked with a script). Coral and teal answers use dark text instead of white.
- **Icons instead of emoji,** so the emotes and status marks look the same on every phone and projector.
- **Accessibility:** visible keyboard focus, arrow-key shop tabs, 44px touch targets, screen-reader announcements for results and purchases, and reduced motion for students who turn animations off.

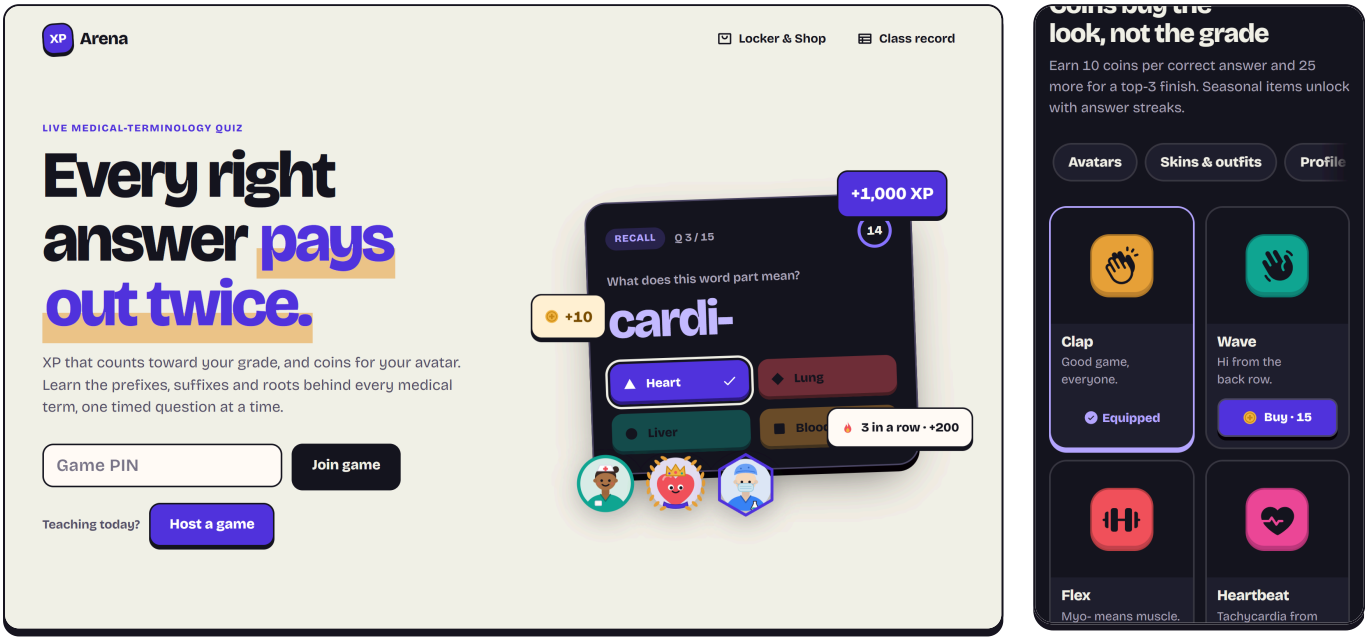


Figure 7. Home page in the new design (left) and the Coin Shop on a phone in dark mode, with the new icon-based emotes (right).

Task 8 Tested a full round and fixed what we found

D David DONE

We played complete 15-question rounds in the browser at phone size (375 × 812), projector size (1280 × 720) and laptop size, in light and dark mode, with the keyboard only, and with reduced motion. Code checks (typecheck, 6 tests, production build) pass. The issues found and how they were handled are listed in section 5.

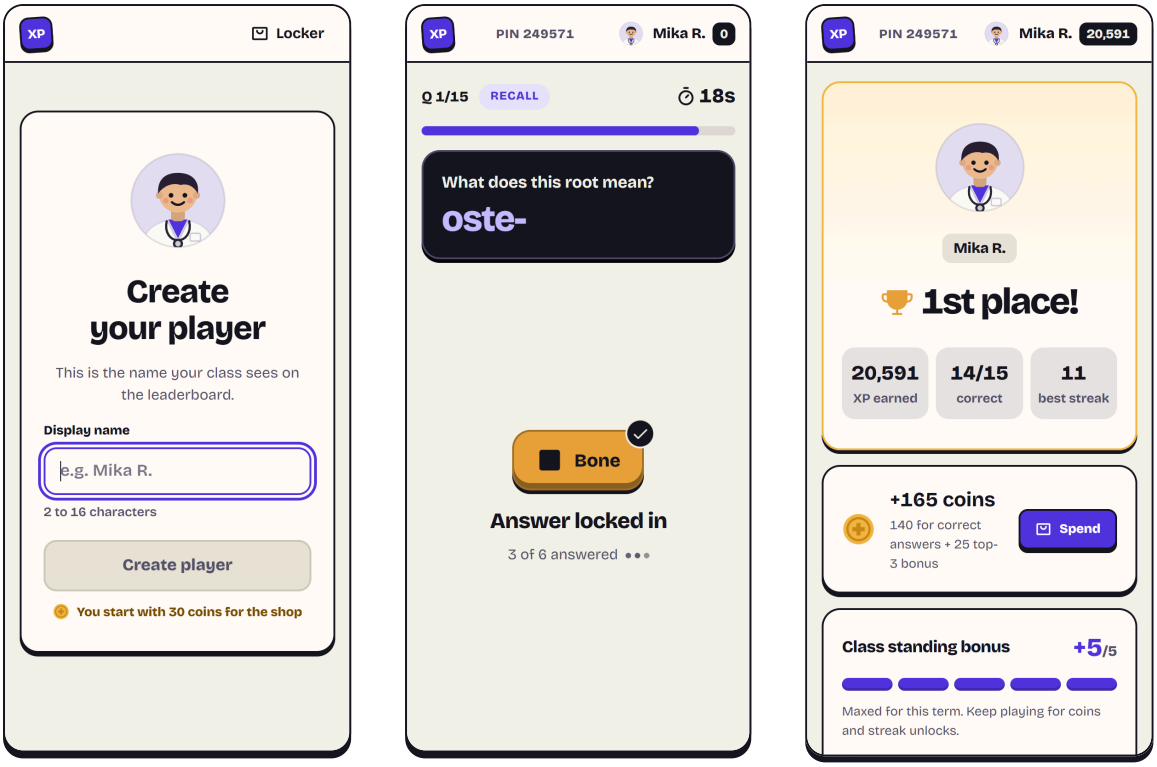


Figure 8. More phone states from the test round: creating a player from the QR link, an answer locked in while others finish, and the end-of-round summary with coins and grade-bonus progress.

5 Remarks / Issues Encountered Today

ISSUE	WHAT WE DID	STATUS
The game server crashed on start because the preview tool and the server both wanted port 5173.	The server now uses its own setting (`API_PORT`, default 3001).	FIXED
The projector showed students the wrong join address (the server's internal port).	The join address now uses the page's own port.	FIXED
The class record overstated accuracy: skipped questions weren't counted.	Accuracy now counts all 15 questions, so skipping counts against the 80% target.	FIXED
White text on the coral and teal answer colors was hard to read (about 3:1 to 3.5:1 contrast, below the 4.5:1 standard).	Switched to dark text on those colors (5.4:1 and 5.9:1), keeping the deck's exact colors.	FIXED
Emoji emotes and marks looked different on each device.	Replaced every emoji with matching icons.	FIXED
On phones, the Locker page ran off the right edge of the screen.	Fixed the layout so the category tabs scroll inside their row.	FIXED

ISSUE	WHAT WE DID	STATUS
On smaller projectors the game PIN wrapped onto two lines and the join address broke in the middle.	Resized the PIN and kept the address together.	FIXED
On the reveal screen, wrong answers were faded so much their text was unreadable; 4th place wrongly got the gold badge.	Wrong answers now use a neutral block with readable text; gold goes only to rank 1.	FIXED
Small display bugs: off-center purchase message, faint "need more coins" buttons, cream background showing in dark mode, muddy title highlight in dark mode.	Each corrected and re-checked in the browser.	FIXED
Not yet tested on real phones over the classroom Wi-Fi.	Planned for the next session. Windows Firewall may ask to allow Node.js the first time.	OPEN
Prototype limits: one shared class record, no teacher login or student accounts, data kept in a single file; restarting the server ends a live round (profiles and coins are kept).	Acceptable for a classroom prototype; to be revisited if the game moves past the prototype stage.	NOTED

Next steps

- Run a test session with real phones on the classroom Wi-Fi.
- Collect feedback from classmates on question difficulty and the 20-second timer.
- Confirm that the 80% accuracy target and the +5 grade cap feel fair after a real round.